

Temporal Integrity Research & Testing Framework

DRAFT ≠ CANON | Independent Execution Framework

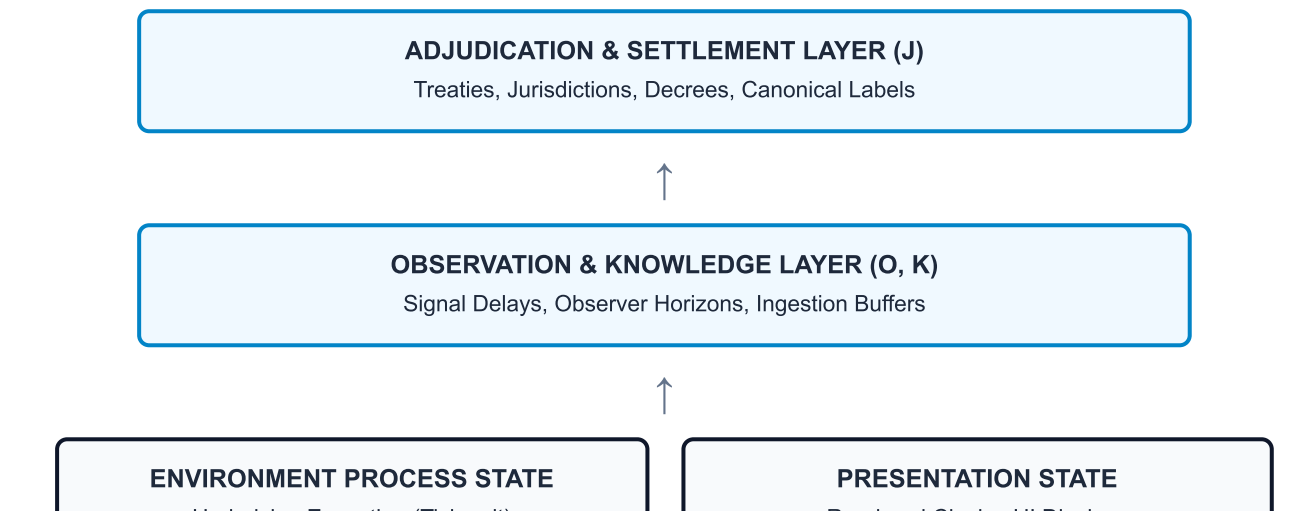
I. Comparative Evaluation of Prior Frameworks

This laboratory design dismantles the fallacy of a universal clock and establishes an append-only event philosophy. To create a fully actionable testing laboratory, operational gaps from prior conceptual models have been resolved into a synthesized target architecture.

Dimension	Prior Analyses (TITF & Grok)	Synthesis & Laboratory Target
Ontology & Units	Deconstructs time into multiple inquiry tracks, a 9-tuple coordinate, and four distinct unit kinds (Duration, Order, Presentation, Settlement).	Combines the four-kind operational split with a precise coordinate tuple, binding each explicitly to reference frames and ruleset versions.
Epistemology & Authority	Identifies that authority is governance, not truth. Emphasizes epistemic labels, test oracles, and fault checkers.	Separates governance from evidentiary certainty, enforced through deterministic, automated test oracles.
System Architecture	Strong operational primitives (Bindings, Intervals) leaning toward traditional distributed systems and temporal databases.	Advances beyond static DAGs into a Versioned Causal Hypergraph capable of modeling active Time-War mechanics.

II. Core Architectural Model: The SOPA Engine

To ensure that neither fictional powers nor system failures collapse reality into an unversioned mush, this laboratory operates on the **SOPA Model** (State, Observation, Presentation, Adjudication).



Mathematical Primitives

The framework operates on strict mathematical primitives to avoid semantic ambiguity:

1. The Event (E): An immutable physical or computational state change. Never contains an intrinsic timestamp.

$$E = \langle UUID, Payload, DomainID, P_env \rangle$$

(Where P_env is the environmental process delta)

2. The Temporal Marker (M): A coordinate location within a specified counting apparatus.

$$M = \langle \tau, U, C_id, F_id, R_v, \sigma^2 \rangle$$

(Where τ is scalar value, U is unit kind, C_id is clock identity, F_id is frame, R_v is ruleset, σ^2 is uncertainty margin)

3. The Temporal Binding (B): The cryptographic/evidentiary link between Event and Marker.

$$B = \langle E_UUID, M, BinderID, Method, EvidenceRef, WitnessSet \rangle$$

(A marker without a verifiable binding is classified as an unauthenticated rumor)

4. The Causal Hypergraph (H):

$$H = (V_e, E_c, T_r)$$

(Where V_e are Event nodes, E_c are directed causal edges, and T_r are versioned transformation functions mapping coordinates across domains)

III. Execution Protocols for Core Scenarios

Protocol 1: The Century Token ($\Omega 100$) Audit Engine

Hypothesis: "Purchasing 100 years of time changes" cannot grant ownership over elapsed physical history; it constitutes an **Asset-Bounded Audit and Settlement License** over an explicitly declared reference interval $[M_start, M_end)$.

- **Step 1 (Corpus Ingestion):** Scope the audit to a closed corpus of documented territories over a defined reference interval.
- **Step 2 (Classification):** Categorize events into operational classes (e.g., Presentation Shifts like `OFFSET_CHANGE`, Execution Alterations like `TICK_RATE_CHANGE`, or Topological Events like `TIMELINE_BRANCH`).
- **Step 3 (Reconciliation):** Compute a Dispute Surface S for contradictory evidence, logging jurisdictional provenance without artificially forcing a merged timeline.

Protocol 2: The Bounded 1-Hour Sandbox

Hypothesis: An isolated 1-hour block is a closed-open interval object $I = [M_0, M_1)$ whose behavior varies across execution environments without invalidating the master reference baseline.

```
Reference Clock R: [00:00 -----> 01:00] (3600s)
Env A (1x Real-Time): [00:00 -----> 01:00] (Local Elapsed: 1h) Env B (10x
Accelerated):[00:00 ---> 10:00] (Local Elapsed: 10h; Rate Shift) Env D (Engine Pause):
[00:00 -> [PAUSED] -> 00:30] (Local Elapsed: 30m) Env E (Event-Driven): [e1, e2 ... e38]
(Local Elapsed: 38 Ticks; 0 Duration) Env F (Fall-Back Fold): [01:00 -> 01:59 -> 01:00 -
> 01:30] (Aliased Identifiers)
```

- **Validation Rule:** Any event occurring within an Environment must be indexed with both its local coordinate and its projection onto the external reference interval.
- **Boundary Crossing:** Events initiating within the interval but concluding outside it must be decomposed into an interior segment, a boundary transition event, and an exterior continuation.

Protocol 3: Cosmological Metric & Origin-of-Light Interventions

Hypothesis: A light-year is strictly a unit of spatial length. "Purchasing light-years" designates the acquisition of a spatial transmission corridor or an observation latency window.

```
ORIGIN INTERVENTION (t0) | v \ / \ AFFECTED / <-- Causal Dependency Cone \ CONE /
(Recalculated State) \ / -----*-----*----- UNAFFECTED
REGION UNAFFECTED REGION (Histories invariant) (Histories invariant)
```

If a speculative capability alters light propagation at cosmic initialization, the engine calculates a **Causal Dependency Light Cone**. Spacetime coordinates outside this light cone remain unmutated. The ledger generates pre-intervention and post-intervention regimes.

IV. Time-War Dynamics & Attack Vector Probes

In an active Time War, combatants do not merely adjust physical history; they attack reference frames, observation pipes, and temporal registries. The lab assesses distinct attack vectors:

Probe ID	Attack Vector	Adversarial Mechanism	Ledger Defense
ADV-01	Temporal Aliasing Injection	Attacker forces a backward clock offset to give two distinct events identical presentation labels.	Pass: System binds distinct sequence IDs and monotonic UTC ticks.
ADV-02	Presentation Hijacking	Inhabitants' displays are updated with a forged "now," while underlying engine ticks continue unaffected.	Pass: SOPA model logs presentation divergence while preserving process counters.
ADV-03	Causal Edge Erasure	Attacker retroactively asserts that Event B preceded Event A to invalidate a dependency.	Pass: Hypergraph validates vector clocks; highlights causal impossibility (cycles).
ADV-04	Rate-Arbitrage Spoofing	Attacker accelerates an environment clock from 1x to 5x to trigger a deadline prematurely.	Pass: Contract validation evaluates duration units, not raw presentation labels.
ADV-05	Latency Horizon Exploitation	Exploiting propagation delay to commit an action at Planet B before news of a treaty signed at Planet A arrives.	Pass: Ingestion tags verify observation horizons, preventing retroactive liability.
ADV-06	Ruleset Obfuscation	Replaying Tick 1000 from engine v1 inside engine v2 where physics constants differ.	Pass: Strict validation gate rejects markers missing exact ruleset_version.

V. Formal Validation Gates & Reconciliation Protocols

Every submitted event claim must pass through eight ordered gates before classification:

1. **Schema Gate:** Verify UUID, Subject, Domain, and Payload completeness.
2. **Binding Gate:** Authenticate BinderID, Witness signatures, and Scale metadata.
3. **Frame/Clock Gate:** Verify declared ClockID exists and is active over target interval.
4. **Mapping Gate:** Confirm conversion functions are explicitly versioned.
5. **Causal Gate:** Evaluate Lamport/Vector clocks; verify DAG acyclicity.
6. **Provenance Gate:** Validate derivation lineage (W3C PROV compliance).
7. **Adjudication Gate:** Verify jurisdictional authority over the specific claim layer.
8. **Classification:** Assign Status [Validated | Conditionally Validated | Disputed | Indeterminate].

Reconciliation Protocol (Query-Driven)

When a dispute arises across territories or timelines, reconciliation is driven by an explicit query rather than forcing a global timeline merge:

1. **Freeze the Query:** State the exact question under review (e.g., "Did Transaction A settle before Treaty B?").
2. **Isolate Domain Contexts:** Extract local timelines and internal orderings.
3. **Execute Versioned Translations:** Apply declared mapping functions. Unmappable intervals become residuals.
4. **Generate Difference Table:** Render comparative findings.

Domain / Party	Claimed Instant	Normalized Ref	Causal Predecessor	Status
Territory Alpha	09:15:00 (v1)	16:15:00Z	E_0 (Gate Open)	Corroborated
Territory Beta	10:15:00 (v2)	16:15:00Z	E_0 (Gate Open)	Corroborated (1 hr offset)
Adv. Interceptor	08:45:00 (v1)	15:45:00Z	None (Orphaned)	Disputed (Causal Anomaly)

VI. The Laboratory Test Harness & Invariant Battery

The laboratory validates model stability by executing a rigorous invariant test battery:

INV-01: Identity Invariance. Coordinate remapping or time-zone shifts must never alter the underlying event UUID.

INV-02: Frame Completeness. Reject timestamps lacking explicit clock, frame, ruleset, and scale.

INV-03: Causal Monotonicity. Prevents backward causal loops unless explicit retrocausality protocol is invoked.

INV-04: Non-Destructive Corrections. Append knowledge corrections; never overwrite historical assertions.

INV-05: Mapping Determinism. Identical inputs to mapping functions yield byte-for-byte reproducible output.

INV-06: Discontinuity Preservation. Gaps and folds stay visible and are flagged, not smoothed out.

INV-07: Uncertainty Conservation. Mathematical translations cannot arbitrarily narrow a declared uncertainty margin.

INV-08: Provenance Lineage. Unbroken derivation chains referencing observers, instruments, and authorities.

INV-09: Branch Isolation. No cross-timeline data leakage without an explicit bridge transaction.

INV-10: Authority Decoupling. Decrees cannot retroactively alter physical sensor logs.

INV-11: Token Boundary Lockdown. Audit rights cannot execute state traversals or edits.

INV-12: SOPA State Separation. Rendered UI clocks do not dictate underlying process execution ticks.

VII. Phased Implementation Roadmap

1. **Phase 0: Decision Register Finalization.** Lock in foundational canon choices (e.g., biological aging rules, reference clock baselines).
2. **Phase 1: Formal Schema Publication.** Publish strict schemas for Event Claim Records and Bindings.
3. **Phase 2: Deterministic Reference Engine Assembly.** Build the in-memory causal hypergraph engine and vector clocks.
4. **Phase 3: Baseline Suite Execution.** Run test suites simulating the 1-Hour Sandbox, Century Token Audit, and Metric Shifts.
5. **Phase 4: Adversarial Red-Teaming.** Inject Time-War attack vectors to validate ledger resilience.
6. **Phase 5: Project Canon Integration.** Promote verified mechanisms into core development logs.